Embedded Computer Vision Final Project Proposal

Phil Orlando
Graduate Student, Department of Electrical Engineering
University of Colorado Boulder

1. My final project will be an assistant for blackjack the card game. It will start with image recognition of cards and when complete, be able to recognize the dealer and player hands and make recommendations to the player by applying statistical likelihood of winning based on the dealers show card and the players cards.

   To make this work, I expect to use blur, canny edge detection, contour finding, and comparison with a template of cards to determine the value. I will take the template picture with the same deck I will use for the project so that the correlation can be as high as possible. The table and camera view will be split like a blackjack table where the dealer cards are on one side of the table, and the player cards are on the other side. By splitting the image, I will determine who has which card.

   Once multiple cards can be recognized from the camera feed, the associated values for blackjack will be taken and summed for the player and read for the dealer. Then a statistical analysis using the players two cards and the dealers single showing card to determine the best option for the player. Then the recommendation for the players action will be displayed.

   Due to the blackjack recommendation not needing to be displayed as fast as autonomous car steering recommendations, I believe 10 frames per second should be enough for a fast response to the player without overburdening the processing with a nearly static screen. Frames per second goals are attached in the MIN, TGT, and OPT matrix.

Outcome	Feature	Throughput performance	Reliability
Minimum	Detects multiple cards values	10 FPS	P>0.8 R>0.8
Target	Determine dealer and player cards along with associated blackjack value	20 FPS	P>0.9 R>0.9
Optimal	Determine recommendation for	30 FPS	P>0.95

	players action and handle additional player cards		R>0.95